

# $\mu\text{Ne}$ 俘获截面

黄新华

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# 理论基础

- 参考1975与1977两篇论文的理论公式利用python代码实现俘获截面计算
- 1975提供了初始版本方法，1977提供了改进版方法，具体细节参考原文了解

P. Vogel, P. K. Haff, V. Akylas, A. Winther, Nucl. Phys. A254 (1975) 445.

P. Vogel, A. Winther, V. Akylas, Phys. Lett. B70 (1977) 39.

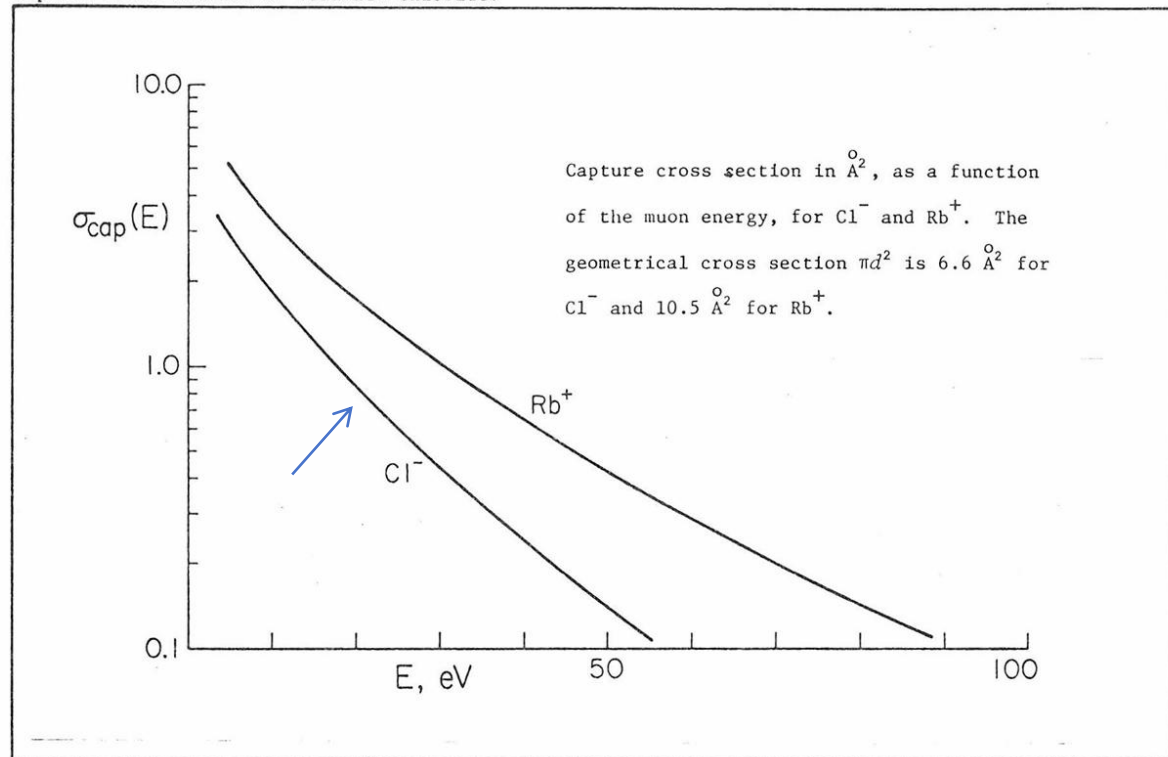
此外V. Akylas的博士论文“CASCADE OF NEGATIVE MUONS IN ATOMS”也提供了计算缪子俘获截面的一些内容

# 代码验证

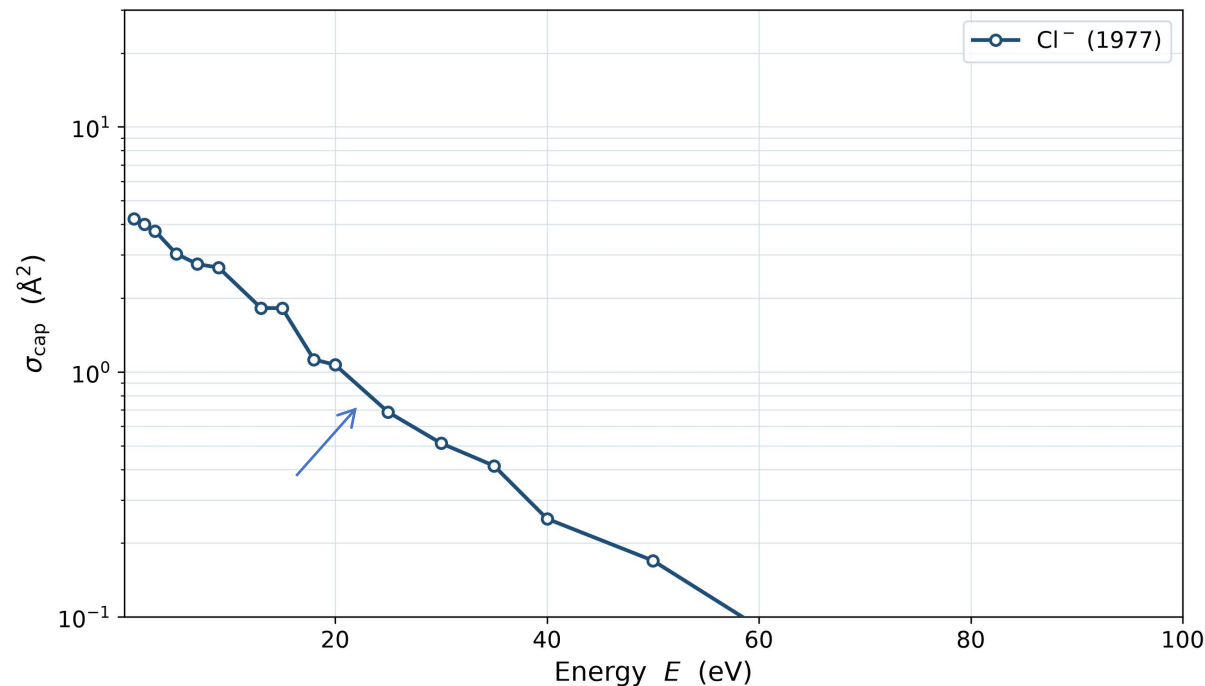
- 用1977论文里面 $\text{Cl}^-$  (氯离子) 的俘获截面图片验证代码对1977方法的实现是成功的(如下两图)

Figure 1.1

Capture cross section of rubidium chloride.



$\text{Cl}^-$  Muon Capture Cross Section (1977)



- 1975论文没有具体的俘获截面计算验证的图，验证代码实现是否成功依靠里面特定参数的交叉验证，在实现的python代码readme文件有详细介绍

# 计算结果

Ne Muon Capture Cross Section (1977)

| E (eV) | $\sigma_{\text{cap}} (\text{\AA}^2)$ | E (eV) | $\sigma_{\text{cap}} (\text{\AA}^2)$ |
|--------|--------------------------------------|--------|--------------------------------------|
| 1      | 16.0118                              | 25     | 0.2405                               |
| 2      | 9.5056                               | 30     | 0.1637                               |
| 3      | 7.6984                               | 35     | 0.1172                               |
| 5      | 4.9607                               | 40     | 0.0817                               |
| 7      | 2.7781                               | 50     | 0.0406                               |
| 9      | 2.0276                               | 60     | 0.0295                               |
| 13     | 1.3819                               | 70     | 0.0204                               |
| 15     | 0.8256                               | 80     | 0.0144                               |
| 18     | 0.5231                               | 90     | 0.0086                               |
| 20     | 0.4088                               | 100    | 0.0065                               |

Ne Muon Capture Cross Section (1975)

| E (eV) | $\sigma_{\text{cap}} (\text{\AA}^2)$ | E (eV) | $\sigma_{\text{cap}} (\text{\AA}^2)$ |
|--------|--------------------------------------|--------|--------------------------------------|
| 1      | 20.5969                              | 25     | 0.0088                               |
| 2      | 14.7239                              | 30     | 0.0041                               |
| 3      | 11.3545                              | 35     | 0.0018                               |
| 5      | 8.4218                               | 40     | 0.0018                               |
| 7      | 7.1192                               | 50     | 0.0008                               |
| 9      | 5.9258                               | 60     | 0.0008                               |
| 13     | 3.8665                               | 70     | 0.0003                               |
| 15     | 1.9927                               | 80     | 0.0003                               |
| 18     | 0.0834                               | 90     | 0.0003                               |
| 20     | 0.0187                               | 100    | 0.0000                               |

# Multielectron effects in capture of antiprotons and muons by helium and neon

James S. Cohen\*

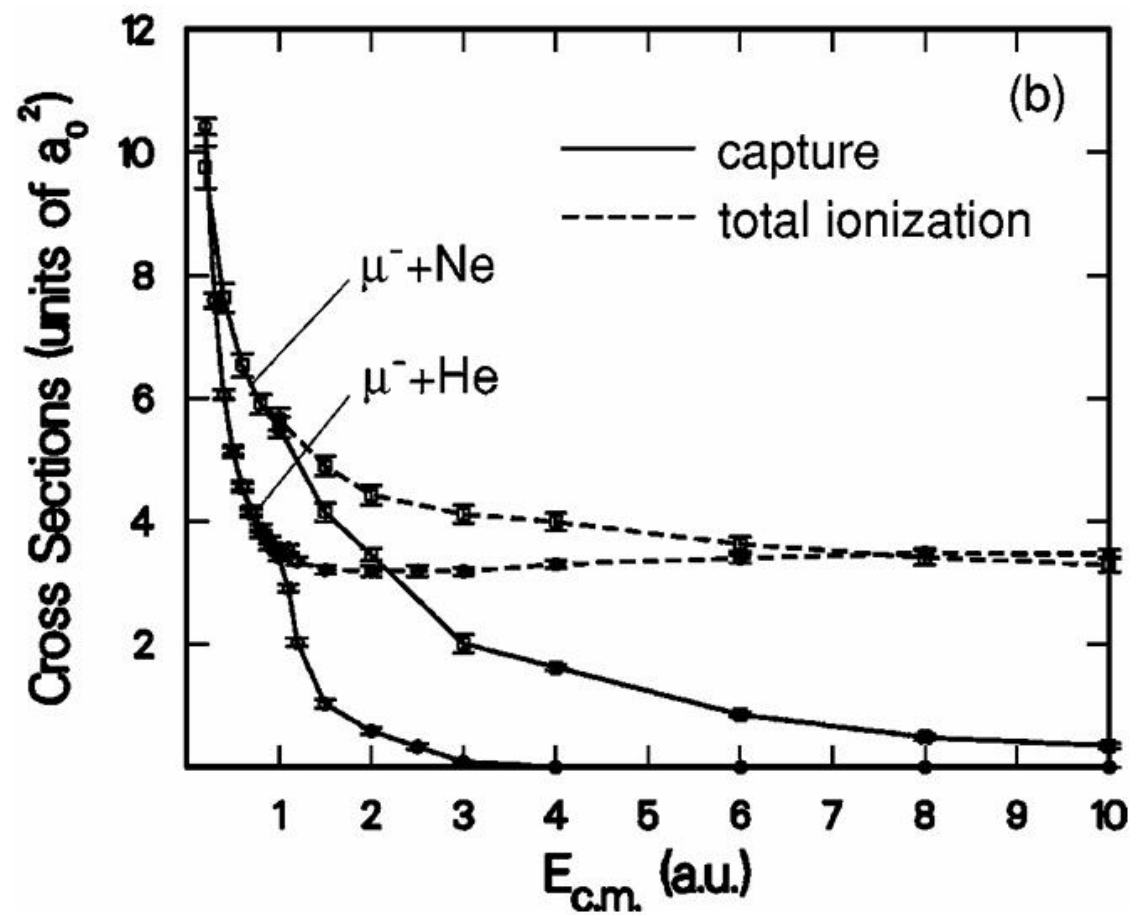
*Theoretical Division, Los Alamos National Laboratory, Los Alamos, New Mexico 87545*

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Antiproton ( $\bar{p}$ ) and negative muon ( $\mu^-$ ) captures by helium and neon atoms are treated using the fermion-molecular-dynamics method, yielding capture cross sections, initial quantum numbers, and ejected-electron energies. The calculated angular-momentum distributions tend to pile up at  $l=n-1$  and are not well fitted by the form  $(2l+1)e^{\alpha l}$ , which is often assumed in experimental analyses. The residual electrons are generally left in a “shake-up” state. When capture is accompanied by multiple ionization, the second and later electrons escape with increasing kinetic energies, a process that is not well described as quasiadiabatic. In a 50:50 mixture of helium and neon, the calculated Ne:He capture ratios are 3.36 for  $\bar{p}$  and 3.69 for  $\mu^-$ . The  $\mu^-$  per-atom capture probability is almost independent of the neon fraction and is in overall agreement with several  $\mu^-$  and  $\pi^-$  experiments; for  $\bar{p}$  there is a rather strong dependence on neon fraction, but there are as yet no experiments. The possible experimental indication of a strong isotope effect on capture of  $\pi^-$  in mixtures of  $^3\text{He}$  and  $^4\text{He}$  is not supported by the present calculation.

PACS number(s): 36.10.-k, 34.10.+x, 25.43.+t, 03.65.Sq

\*采用费米子分子动力学（FMD）方法，研究了负μ子在氖（Ne）原子中的捕获过程，给出了μNe俘获截面与总电离截面数据



| $E_{c.m.}$ (a.u.) | $\sigma_{\text{capt}}$ (units of $a_0^2$ ) | $\sigma_{\text{tot}}$ (units of $a_0^2$ ) |
|-------------------|--------------------------------------------|-------------------------------------------|
| 0.01              | $36.19 \pm 2.20$                           | $24.88 \pm 2.39$                          |
| 0.10              | $12.82 \pm 0.51$                           | $12.44 \pm 0.50$                          |
| 0.20              | $9.75 \pm 0.35$                            | $9.75 \pm 0.35$                           |
| 0.40              | $7.63 \pm 0.23$                            | $7.63 \pm 0.23$                           |
| 0.60              | $6.53 \pm 0.19$                            | $6.53 \pm 0.19$                           |
| 0.80              | $5.91 \pm 0.16$                            | $5.91 \pm 0.16$                           |
| 1.00              | $5.53 \pm 0.17$                            | $5.66 \pm 0.17$                           |
| 1.50              | $4.15 \pm 0.15$                            | $4.90 \pm 0.16$                           |
| 2.00              | $3.46 \pm 0.10$                            | $4.43 \pm 0.16$                           |
| 3.00              | $2.01 \pm 0.15$                            | $4.12 \pm 0.15$                           |
| 4.00              | $1.63 \pm 0.05$                            | $3.99 \pm 0.15$                           |
| 6.00              | $0.86 \pm 0.03$                            | $3.63 \pm 0.13$                           |
| 8.00              | $0.49 \pm 0.04$                            | $3.41 \pm 0.12$                           |
| 10.00             | $0.35 \pm 0.04$                            | $3.29 \pm 0.12$                           |
| 12.00             | $0.22 \pm 0.03$                            | $3.20 \pm 0.12$                           |
| 15.00             | $0.10 \pm 0.02$                            | $2.92 \pm 0.10$                           |
| 18.00             | $0.03 \pm 0.01$                            | $2.88 \pm 0.10$                           |
| 21.00             | $0.03 \pm 0.01$                            | $2.83 \pm 0.10$                           |
| 25.00             | $0.01 \pm 0.01$                            | $2.62 \pm 0.10$                           |
| 30.00             |                                            | $2.52 \pm 0.10$                           |

$$\tau_{\text{cap}}^{\text{th}} \simeq \frac{1}{n\sigma_{\text{cap}}^{\text{eff}}v_{\text{th}}},$$

| $E_{\text{c.m.}}$ (a.u.) | $\sigma_{\text{capt}}$ (units of $a_0^2$ )              | $\sigma_{\text{tot}}$ (units of $a_0^2$ ) |
|--------------------------|---------------------------------------------------------|-------------------------------------------|
| 0.01                     | $1 \text{ a.u.} = 27.21 \text{ eV}$<br>$36.19 \pm 2.20$ | $24.88 \pm 2.39$                          |

- $E = 0.27 \text{ eV}$  时的速度  $v = \sqrt{2E/m_\mu} \approx 2.1 \times 10^4 \text{ m/s}$
- 常数  $\sigma v \approx 1.01 \times 10^{-19} \times 2.1 \times 10^4 \approx 2.1 \times 10^{-15} \text{ m}^3/\text{s}$

\*俘获时间满足预测的0.1-1微秒

$$\tau_{\text{base}} \approx \frac{1}{n \cdot (\sigma v)}$$

取2026年论文的密度  $n = 10^{22} \text{ m}^{-3}$ :

$$\tau_{\text{base}} \approx \frac{1}{10^{22} \times 2.1 \times 10^{-15}} \approx 4.8 \times 10^{-8} \text{ s} \approx \mathbf{48 \text{ ns}}$$

$$\tau_{\text{cap}} = 48 \text{ ns} \times (5 \sim 10) \approx \mathbf{0.24 \sim 0.48 \mu\text{s}}$$

# $\mu\text{Ne}$ 俘获级联完成后电荷态 情况

黄新华

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## Muonic Atoms with Vacant Electron Shells

R. Bacher, D. Gotta, and L. M. Simons

*Institut für Kernphysik, Kernforschungszentrum Karlsruhe, Karlsruhe, Federal Republic of Germany, and Institut für Experimentelle Kernphysik, Universität Karlsruhe, Karlsruhe, Federal Republic of Germany*

and

J. Missimer and Nimai C. Mukhopadhyay<sup>(a)</sup>

*Theory Group, Schweizerisches Institut für Nuklearforschung, CH-5234 Villigen, Switzerland*

(Received 31 January 1985)

We show that the cascade in muonic atoms with  $Z < 20$  ejects sufficient atomic electrons to ionize an isolated muonic atom completely. In gases, the rates with which electrons refill the atomic shell can be accurately deduced from measured and calculated electron transfer cross sections. Thus, we can conclude that completely ionized muonic atoms can be prepared in gases, and that they remain isolated for long enough times at attainable pressures to facilitate studies of fundamental interactions in muonic atoms.

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\*对于 $Z < 12$ 的原子（如Ne）， $\mu$ 子级联过程中，电子几乎完全被电离，电子在 $\mu$ 子到达 $n = 5$ 时仍被束缚的概率  $< 0.25$ 。对于 $12 \leq Z < 20$ 的原子（如Ar），即使初始电子壳层完整，最终完全电离的概率也较高（ $< 0.6$  的电子残留概率）。因此，在轻 $\mu$ 子原子中， $\mu$ 子级联过程极有可能使其完全电离，特别是在低气压气体环境中。

\*提供了Ne $10^+$ 的电子转移截面

\*通过控制气压，可以调节电子再填充速率，从而改变电荷态。

